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!pip -q install pandas numpy scikit-learn joblib

import os
import re
import shutil
import joblib
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report,
    confusion_matrix
)

os.makedirs("smarthire/data/raw", exist_ok=True)
os.makedirs("smarthire/data/interim", exist_ok=True)
os.makedirs("smarthire/data/processed", exist_ok=True)
os.makedirs("smarthire/models", exist_ok=True)

print("Project folders ready.")

csv_files = [
    f for f in os.listdir(".")
    if f.lower().endswith(".csv")
]
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]

if len(csv_files) == 0:
    raise FileNotFoundError(
        "No CSV file found. Please upload your Resume dataset."
    )

print("CSV files found:")

for f in csv_files:
    print(" -", f)

resume_file = csv_files[0]

print("\nUsing:", resume_file)


# =====
# 4. COPY DATASET
# =====

destination = "smarthire/data/raw/Resume.csv"

shutil.copy(
    resume_file,
    destination
)

print(
    "\nDataset copied to:",
    destination
)

df = pd.read_csv(
    destination
)

print("\nDataset shape:")
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print(df.shape)

print("\nDataset columns:")
print(df.columns.tolist())

print("\nFirst 5 records:")
display(df.head())

required_columns = [
    "Resume_str",
    "Category"
]

for column in required_columns:
    if column not in df.columns:
        raise ValueError(
            f"Required column '{column}' not found."
        )

print(
    "\nRequired columns found successfully."
)

data = df[
    [
        "Resume_str",
        "Category"
    ]
].copy()

data.columns = [
    "resume",
    "category"
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]

data = data.dropna(
    subset=[
        "resume",
        "category"
    ]
)

data["resume"] = data["resume"].astype(str)

data["category"] = data["category"].astype(str)

def clean_text(text):

    text = text.lower()

    text = re.sub(
        r"http\S+|www\S+",
        " ",
        text
    )

    text = re.sub(
        r"\S+@\S+",
        " ",
        text
    )

    text = re.sub(
        r"^[a-zA-Z0-9+#\.\s]",
        " ",
        text
    )

    text = re.sub(
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        r"\s+",
        " ",
        text
    )

    return text.strip()

data["resume"] = data["resume"].apply(
    clean_text
)

data = data.drop_duplicates(
    subset=["resume"]
)

data = data[
    data["resume"].str.len() > 30
]

print("\n=====")
print("DATASET INFORMATION")
print("=====")

print(
    "Total resumes:",
    len(data)
)

print(
    "Number of job categories:",
    data["category"].nunique()
)

print("\nJob categories:")

print(
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data["category"].value_counts()
)

processed_path = (
    "smarthire/data/processed/resume_processed.csv"
)

data.to_csv(
    processed_path,
    index=False
)

print(
    "\nProcessed dataset saved:"
)

print(processed_path)


X = data["resume"]
y = data["category"]


X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

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print("\n=====")
print("TRAIN TEST SPLIT")
print("=====")

print(
    "Training samples:",
    len(X_train)
)

print(
    "Testing samples:",
    len(X_test)
)

model = Pipeline([

    (
        "tfidf",
        TfidfVectorizer(

            stop_words="english",

            max_features=15000,

            ngram_range=(1, 2),

            sublinear_tf=True,

            min_df=1
        )
    ),
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(
    "classifier",
    LogisticRegression(
        max_iter=2000,
        C=2.0,
        class_weight="balanced"
    )
)

]

print("\n=====")
print("TRAINING MODEL")
print("=====")

model.fit(
    X_train,
    y_train
)

print(
    "Training completed successfully!"
)

y_pred = model.predict(
    X_test
)

accuracy = accuracy_score(
    y_test,
    y_pred
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)

precision = precision_score(
    y_test,
    y_pred,
    average="weighted",
    zero_division=0
)

recall = recall_score(
    y_test,
    y_pred,
    average="weighted",
    zero_division=0
)

f1 = f1_score(
    y_test,
    y_pred,
    average="weighted",
    zero_division=0
)

print("\n=====")
print("SMART HIRE MODEL RESULTS")
print("=====")

print(
    f"Accuracy : {accuracy:.4f}"
)

print(
    f"Precision : {precision:.4f}"
)

print(
    f"Recall : {recall:.4f}"
)
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print(
    f"F1 Score : {f1:.4f}"
)

print("\n=====")
print("CLASSIFICATION REPORT")
print("=====")

print(
    classification_report(
        y_test,
        y_pred,
        zero_division=0
    )
)

cm = confusion_matrix(
    y_test,
    y_pred
)

print("\n=====")
print("CONFUSION MATRIX")
print("=====")

print(cm)

model_path = (
    "smarthire/models/classifier.pkl"
)

joblib.dump(
    model,
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        model_path
    )

    print(
        "\nModel saved successfully:"
    )

    print(model_path)


sample_resume = """

Python developer with experience in
machine learning, data analysis,
Pandas, NumPy, Scikit-learn,
SQL, artificial intelligence,
data visualization and Python.

"""

sample_resume = clean_text(
    sample_resume
)

prediction = model.predict(
    [sample_resume]
)

print("\n=====")
print("SAMPLE RESUME PREDICTION")
print("=====")

print(
    "Predicted Category:",
    prediction[0]
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)

probabilities = model.predict_proba(
    [sample_resume]
)[0]

classes = model.classes_

probability_df = pd.DataFrame({
    "Category": classes,
    "Probability": probabilities
})

probability_df = probability_df.sort_values(
    by="Probability",
    ascending=False
)

print("\nTop 5 predicted categories:")

display(
    probability_df.head(5)
)

print("\n=====")
print("SMART HIRE CLASSIFIER COMPLETED")
print("=====")

print(
    "Model file:",
    model_path
)
```

```
print(  
    "Processed data:",  
    processed_path  
)
```



Project folders ready.

CSV files found:

- Resume.csv
- Resume (1).csv

Using: Resume.csv

Dataset copied to: smarthire/data/raw/Resume.csv

Dataset shape:

(2484, 4)

Dataset columns:

['ID', 'Resume_str', 'Resume_html', 'Category']

First 5 records:

	ID	Resume_str	Resume_html	Category	
0	16852973	HR ADMINISTRATOR/MARKETING ASSOCIATE\...	<div class="fontsize fontface vmargins hmargin...	HR	
1	22323967	HR SPECIALIST, US HR OPERATIONS ...	<div class="fontsize fontface vmargins hmargin...	HR	
2	33176873	HR DIRECTOR Summary Over 2...	<div class="fontsize fontface vmargins hmargin...	HR	
3	27018550	HR SPECIALIST Summary Dedic...	<div class="fontsize fontface vmargins hmargin...	HR	
4	17812897	HR MANAGER Skill Highlights ...	<div class="fontsize fontface vmargins hmargin...	HR	

Required columns found successfully.

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DATASET INFORMATION

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Total resumes: 2481

Number of job categories: 24

Job categories:

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category
INFORMATION-TECHNOLOGY    120
BUSINESS-DEVELOPMENT      119
ADVOCATE                   118
CHEF                       118
ENGINEERING                118
ACCOUNTANT                 118
FINANCE                    117
FITNESS                    117
SALES                      116
AVIATION                   116
BANKING                    115
HEALTHCARE                 115
CONSULTANT                 115
CONSTRUCTION               112
PUBLIC-RELATIONS           111
HR                          110
DESIGNER                   107
ARTS                       103
TEACHER                    102
APPAREL                    97
DIGITAL-MEDIA              96
AGRICULTURE                 63
AUTOMOBILE                 36
BPO                         22

```

Name: count, dtype: int64

Processed dataset saved:
smarthire/data/processed/resume_processed.csv

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TRAIN TEST SPLIT

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Training samples: 1984

Testing samples: 497

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TRAINING MODEL

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Training completed successfully!

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SMART HIRE MODEL RESULTS


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Accuracy : 0.6761
Precision : 0.6926
Recall    : 0.6761
F1 Score  : 0.6609

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CLASSIFICATION REPORT
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	precision	recall	f1-score	support
ACCOUNTANT	0.56	0.83	0.67	24
ADVOCATE	0.48	0.54	0.51	24
AGRICULTURE	0.88	0.54	0.67	13
APPAREL	0.60	0.16	0.25	19
ARTS	0.57	0.38	0.46	21
AUTOMOBILE	0.80	0.57	0.67	7
AVIATION	0.90	0.78	0.84	23
BANKING	0.84	0.70	0.76	23
BPO	0.00	0.00	0.00	4
BUSINESS-DEVELOPMENT	0.58	0.88	0.70	24
CHEF	0.85	0.71	0.77	24
CONSTRUCTION	0.75	0.82	0.78	22
CONSULTANT	0.71	0.22	0.33	23
DESIGNER	0.83	0.90	0.86	21
DIGITAL-MEDIA	0.79	0.58	0.67	19
ENGINEERING	0.74	0.71	0.72	24
FINANCE	0.76	0.54	0.63	24
FITNESS	0.85	0.71	0.77	24
HEALTHCARE	0.65	0.65	0.65	23
HR	0.77	0.91	0.83	22
INFORMATION-TECHNOLOGY	0.71	1.00	0.83	24
PUBLIC-RELATIONS	0.55	0.77	0.64	22
SALES	0.43	0.65	0.52	23
TEACHER	0.60	0.90	0.72	20
accuracy				0.68
macro avg				0.64
weighted avg				0.66